

Menu-Driven C Program — Digit Operations

1. Problem Understanding

The program takes a **positive integer** and displays a menu with three operations:

- 1. Sum of squares of digits** — using a while loop.
- 2. Sum of cubes of digits** — using a for loop.
- 3. Difference between sum of cubes and sum of squares** — using a do-while loop.

For example, for 123:

- Sum of squares = $1^2 + 2^2 + 3^2 = 14$
 - Sum of cubes = $1^3 + 2^3 + 3^3 = 36$
 - Difference = $36 - 14 = 22$
-

2. Algorithm

Case 1 — Sum of Squares

1. Start.
2. Input a positive integer num.
3. Set sum_square = 0.
4. Copy num into temp.
5. Using a while loop:
 - Extract the last digit using $\text{digit} = \text{temp} \% 10$.
 - Add $\text{digit} \times \text{digit}$ to sum_square.
 - Remove the last digit using $\text{temp} = \text{temp} / 10$.
6. Display sum_square.
7. Stop.

Case 2 — Sum of Cubes

1. Start.
2. Input a positive integer num.
3. Set sum_cube = 0.
4. Using a for loop, extract each digit.
5. Add digit × digit × digit to sum_cube.
6. Display sum_cube.
7. Stop.

Case 3 — Difference

1. Start.
2. Input a positive integer num.
3. Set sum_square = 0 and sum_cube = 0.
4. Using a do-while loop, extract every digit.
5. Calculate both its square and cube and add them to their respective sums.
6. Calculate:
difference = sum_cube - sum_square
7. Display difference.
8. Stop.

3. Test Case Table

	TC Input Number	Choice	Expected Output	Purpose
1	123	1	Sum of squares = 14	Given example
2	123	2	Sum of cubes = 36	Given example
3	123	3	Difference = 22	Given example
4	100	1	Sum of squares = 1	Contains zeros
5	100	2	Sum of cubes = 1	Contains zeros
6	100	3	Difference = 0	Difference testing
7	9	1	Sum of squares = 81	Single digit
8	9	2	Sum of cubes = 729	Single digit

TC	Input	Number Choice	Expected Output	Purpose
9	9	3	Difference = 648	Single digit
10	123	4	Invalid choice	Invalid menu choice
11	0	1	Invalid input	Non-positive input

4.C Program

```
#include <stdio.h>
```

```
int main(void)
{
    int num, choice;
    int temp, digit;
    int sum_square, sum_cube, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    /* Check whether the input is positive */
    if (num <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }

    /* Display menu */
    printf("\n--- MENU ---\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference between sum of cubes and sum of squares (do-while loop)\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            sum_square = 0;
            temp = num;

            while (temp > 0)
            {
                digit = temp % 10;
                sum_square += digit * digit;
                temp /= 10;
            }

            printf("Sum of squares of digits = %d\n", sum_square);
            break;
```

```

case 2:
    sum_cube = 0;

    for (temp = num; temp > 0; temp /= 10)
    {
        digit = temp % 10;
        sum_cube += digit * digit * digit;
    }

    printf("Sum of cubes of digits = %d\n", sum_cube);
    break;

case 3:
    sum_square = 0;
    sum_cube = 0;
    temp = num;

    do
    {
        digit = temp % 10;

        sum_square += digit * digit;
        sum_cube += digit * digit * digit;

        temp /= 10;
    } while (temp > 0);

    difference = sum_cube - sum_square;

    printf("Difference = %d\n", difference);
    break;

default:
    printf("Invalid choice! Please choose 1, 2, or 3.\n");
}

return 0;
}

```

5. Compilation

Save the program as:

digit_menu.c

Compile using CC:

cc digit_menu.c

Run:

./a.out

The program was compiled successfully with GCC using the above command.

6. Execution and Testing

Test Case 1 — Sum of Squares

Input:

Enter a positive integer: 123

--- MENU ---

1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)

Enter your choice: 1

Output:

Sum of squares of digits = 14

Status: PASS 

Test Case 2 — Sum of Cubes

Input:

Enter a positive integer: 123

Enter your choice: 2

Output:

Sum of cubes of digits = 36

Status: PASS 

Test Case 3 — Difference

Input:

Enter a positive integer: 123

Enter your choice: 3

Output:

Difference = 22

Status: PASS 

```
└─$ ./a.out
Enter a positive integer: 123

— MENU —
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 1
Sum of squares of digits = 14

(kali@kali)-[~/demo1]
└─$ ./a.out
Enter a positive integer: 123

— MENU —
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 2
Sum of cubes of digits = 36

(kali@kali)-[~/demo1]
└─$ ./a.out
Enter a positive integer: 123

— MENU —
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 3
Difference = 22
```

7. Final Test Result

TC	Choice	Result	Status
1	1	14	✓ PASS
2	2	36	✓ PASS
3	3	22	✓ PASS
4	1	1	✓ PASS
5	2	1	✓ PASS
6	3	0	✓ PASS
7	1	81	✓ PASS
8	2	729	✓ PASS
9	3	648	✓ PASS
10	4	Invalid choice	✓ PASS
11	1	Invalid input	✓ PASS

Conclusion

The menu-driven C program was **successfully compiled, executed, and tested**. The switch-case correctly selects the required operation, while the three operations demonstrate the use of **while, for, and do-while loops** respectively. All prepared test cases produced the expected results